12 V RGB LIGHTING CARRIER

Factory-fitted carrier components; plug-in Arduino Nano 33 IoT supplied separately

INPUT PROTECTION

F1
6.3 A / 125 V fast

Q4
AOD4185 / reverse polarity

VIN_RAW

VIN_FUSED

REV_GATE

D1
12 V gate clamp

J1
12 V IN / centre +

C1
470u / 35 V

D2
SMBJ15A

R13
10k

Reverse polarity PMOS + input TVS; nominal 9 V rail feeds drivers.
D3 prevents USB-powered Nano VIN from energising the control rail.

9 V CONTROL SUPPLY / USB ISOLATION

U3
L78M09 / 9 V

IN

GND

OUT

D3
SS14 / USB isolation

C2
330n / 50 V

C3
10u / 25 V

C4
100n / 50 V

R14
1k8 / minimum load

V12

V9

NANO_VIN

FOUR TOP-ENTRY RJ11 OUTPUTS

160 in at 14.4 W/m = 4.88 A. Four equal sections = 1.22 A per port.
Each section: 40 in / 1.016 m. Maximum per port: 1.5 A / 1.25 m.
All four ports share the same RGB colour and brightness.

FOUR PARALLEL RGB OUTPUTS

F2
3.15 A / 125 V fast

PORT1_12V

+12.V

R

G

B

J4
RJ11 vertical / + R G B

F3
3.15 A / 125 V fast

PORT2_12V

J5
RJ11 vertical / + R G B

F4
3.15 A / 125 V fast

PORT3_12V

J6
RJ11 vertical / + R G B

F5
3.15 A / 125 V fast

PORT4_12V

J7
RJ11 vertical / + R G B

J4–J7: Wuerth 615004144121, top entry.
6 positions, contacts 2 (+12 V), 3 (R), 4 (G), 5 (B).
Pins 1 and 6 have no contacts.

PLUG-IN NANO 33 IoT

USB connector at this end

J2

D13_/_SCK

3V3

AREF

A0

A1

A2

A3

A4_/_SDA

A5_/_SCL

A6

A7

5V_/_VUSB

RESET

GND

VIN

J3

D12_/_CIPO

D11_/_COPI

D10

D9

D8

D7

D6_/_PWM_B

D5_/_PWM_G

D4

D3_/_PWM_R

D2

GND

RESET

RX

TX

Nano left / USB up

Nano right / USB up

J2/J3: 15 ways each, 2.54 mm pitch.
Socket rows 15.24 mm apart on PCB.
Only D3, D5, D6, VIN and both GND
pins connect to the carrier.

3.3 V PWM INPUTS

HIGH = light on

R1
1k

R2
1k

R3
1k

PWM_R

PWM_G

PWM_B

IN_R

IN_G

IN_B

R4
100k

R5
100k

R6
100k

U1
UCC27524A

U2
UCC27524A

ENA

VDD

ENB

INA

INB

OUTA

OUTB

GND

DRIVE_R

DRIVE_G

DRIVE_B

C5
100n / 50 V

C6
4.7u / 50 V

C7
100n / 50 V

C8
4.7u / 50 V

Input tolerance is independent of driver VDD.
U2 channel B is disabled; OUTB is unconnected.

LOW-SIDE RGB SWITCHES

D4
SS54 / cable clamp

D5
SS54 / cable clamp

D6
SS54 / cable clamp

RED

GREEN

BLUE

R7
33R

R8
33R

R9
33R

GATE_R

GATE_G

GATE_B

Q1
AOD4184A

Q2
AOD4184A

Q3
AOD4184A

R10
100k

R11
100k

R12
100k

GND

COMMON RETURN

ASSEMBLY / USE

H1
M3 mounting

H2
M3 mounting

H3
M3 mounting

H4
M3 mounting

50 x 100 mm carrier: all components factory fitted. Nano USB and barrel input face the same short edge.
Nano plugs into J2/J3 with USB toward pin 1. Input: regulated 12 V DC, centre positive, 5.5 x 2.1 mm.
Branch fuses protect against short circuits; they do not enforce the 1.5 A connector limit.

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Title: 12 V RGB lighting carrier – Nano 33 IoT

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